

Quantitative Discovery of Qualitative Information: A General Purpose Document Clustering Methodology

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joint work with
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(talk at the Northeast Methodology Program, New York University, 4/17/09)

The Problem: Discovery from Unstructured Text

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- (We analyze text; our methods apply more generally)

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- That we think of all this as astonishing ... is astonishing

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 - (Perhaps true by definition in unsupervised learning: If we knew the DGP, we wouldn't be at the discovery stage.)

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- **But how to choose from an enormous list of clusterings?**

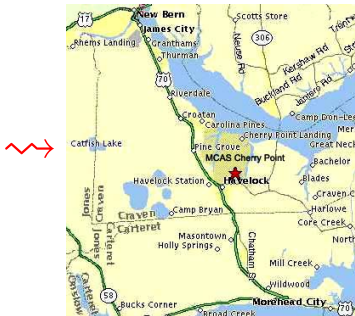
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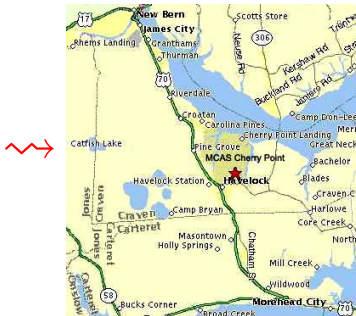
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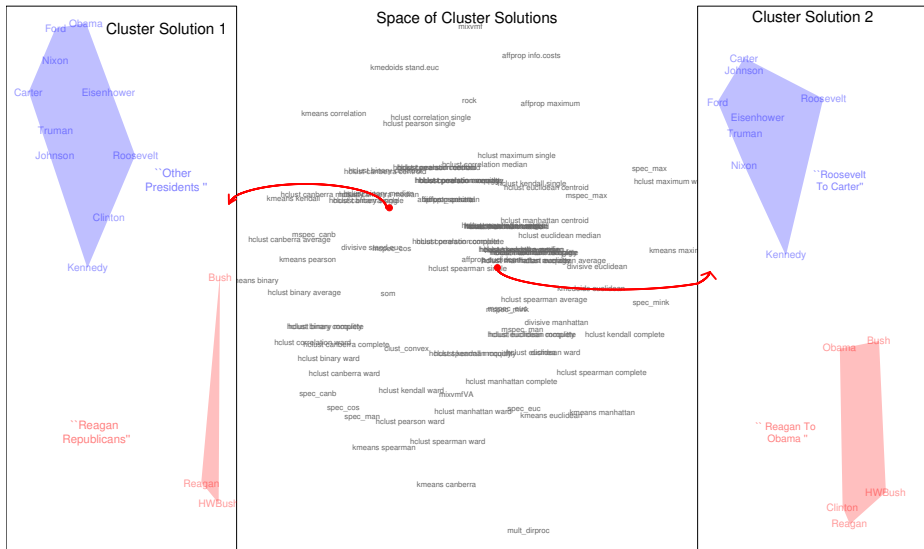
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↪ meaning revealed through a *geography* of clusterings

Many Thousands of Clusterings, Sorted & Organized

You choose one (or more), based on insight, discovery, useful information,...



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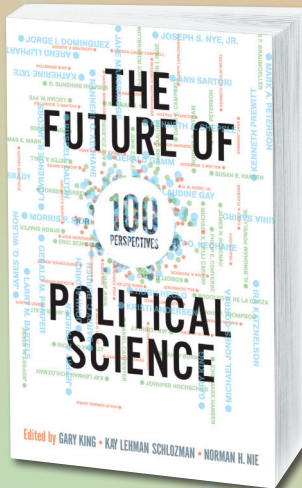
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- ↪ **Only one measure satisfies all three** (the “variation of information”)



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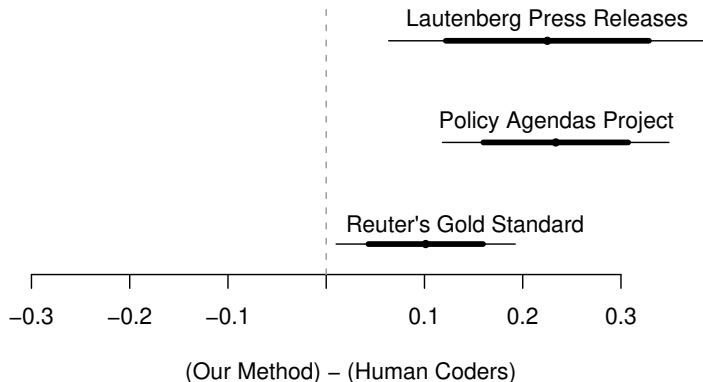
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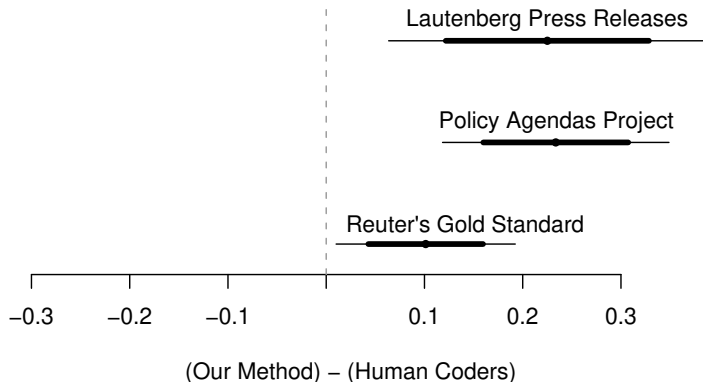
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Lautenberg: 200 Senate Press Releases (appropriations, economy, education, tax, veterans, ...)

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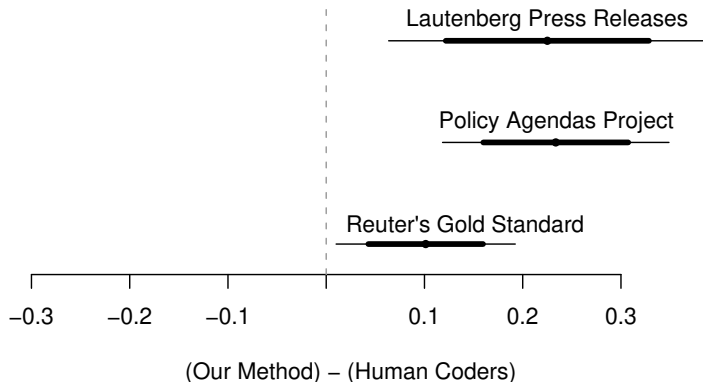
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Policy Agendas: 213 quasi-sentences from Bush's State of the Union (agriculture, banking & commerce, civil rights/liberties, defense, ...)

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Reuter's: financial news (trade, earnings, copper, gold, coffee, ...); "gold standard" for supervised learning studies

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 - ~→ Perhaps this is what it means to be a member of a political party in the U.S.?

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- For each: created 2 clusterings from each of 3 methods, including ours
- Created info packet on each clustering (for each cluster: exemplar document, automated content summary)
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“Genetic testing”:

Our Method 1 → {Our Method 2, K-Means 1, K-means 2} → Dir Proc. 1 → Dir Proc. 2

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- Evaluating new individual methods: (1) distance from existing methods and their averages, (2) usefulness of discoveries in given data sets.

For more information:

<http://GKing.Harvard.edu>